

JASWANTH MADA

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SUMMARY

- Applied AI / Machine Learning Engineer with **4+ years of hands-on experience** delivering ML systems into production, working across model development, evaluation, and backend integration.
- Currently working at **Morgan Stanley** on **LLM and RAG-based internal platforms** using **Python, AWS, and CI/CD**, focused on retrieval quality, prompt behavior control, runtime performance.
- Hands-on experience with **Agentic AI** building multi-agent systems using **CrewAI** with **ReAct** and **Planning** reasoning patterns, role-based agent crews, memory management, guardrails, structured outputs, and human-in-the-loop controls.
- Strong at taking models from **training notebooks to monitored, API-backed systems**, operating across **ML, backend engineering, and cloud infrastructure**.

EXPERIENCE

MORGAN STANLEY | Applied AI Engineer (LLM / GenAI Platform)

Jan 2025 - Current

- Engineered an internal **LLM-powered** document Q&A system using **Python, LangChain, and Pinecone**, reducing time spent on policy and technical document lookups by **35%** for research and engineering users.
- Implemented a scalable document ingestion and chunking pipeline on **AWS (S3, Lambda)** to process and refresh **100K+ pages** of PDFs and markdown content with consistent retrieval behavior.
- Optimized prompt templates, retrieval depth, and citation logic using **OpenAI APIs and prompt versioning**, cutting irrelevant or partially grounded responses by **25%** in internal validation.
- Automated testing and deployment workflows with **GitHub Actions and pytest**, validating prompt changes and retrieval outputs and lowering release-related issues by **30%**.
- Instrumented latency, token usage, and error metrics using **CloudWatch and Python logging**, keeping average response time under **3 seconds** while managing inference cost during peak usage.

SAGE SOFTTECH | Machine Learning Engineer

Mar 2021 - Jul 2023

- Trained an anomaly detection model using **PyTorch autoencoders** on transactional and behavioral data, improving rare-event recall by **20%** compared to prior static rule thresholds.
- Developed fraud classification models in **TensorFlow and scikit-learn** with class-imbalance handling and probability calibration, reducing false positives by **15%** in downstream review workflows.
- Engineered feature sets in **Python and SQL** including rolling-window counts, velocity ratios, and behavioral deltas, supporting stable training on datasets with **8M+ records**.
- Operationalized batch training and inference pipelines on **AWS (S3, EC2)**, enabling scheduled fraud scoring jobs consumed by risk and operations teams.
- Deployed model scoring services as **Python REST APIs on AWS EC2**, maintaining average inference latency under **200 ms** for application-side decisioning.

TECHNICAL SKILLS

Programming & Core ML: Python, NumPy, Pandas, scikit-learn, PyTorch, Supervised & Unsupervised Learning, Anomaly Detection, Fraud Classification, Feature Engineering, Model Evaluation

Agentic AI & Orchestration: AI Agent Architectures, CrewAI, Multi-Agent Systems, ReAct Pattern, Planning Pattern, Tool Use, Custom Tool Development, Agent Memory Management, Guardrails & Output Validation, Human-In-The-Loop, Agent Callbacks, Hierarchical & Sequential Flows, Pydantic Structured Outputs.

LLMs & Retrieval Systems: LLM Integration, Retrieval-Augmented Generation (RAG), LangChain, FAISS, Pinecone, Prompt Engineering, Prompt Versioning, Retrieval Tuning, Citation Control, Embedding-Based Search

Data & Modeling Practices: Data Preprocessing, Rolling-Window Aggregations, Class Imbalance Handling, Model Drift Analysis, Offline Validation

Backend & APIs: Python REST APIs, Inference Services, Application-Level Authorization, API Integration,

Cloud & Deployment: AWS (S3, EC2, Lambda, IAM, CloudWatch), Batch Inference Pipelines

DevOps & CI/CD: Git, GitHub Actions, pytest, Automated Testing, Deployment Pipelines, Version Control for Models & Prompts, Model Performance Monitoring, Cost Monitoring, Production Debugging

PROJECTS

Agentic Content Writing System | Python, Crew AI, OpenAI API, Pydantic

- Built a production grade **Multi Agent content writing system**, evolving from a single agent baseline to a fully orchestrated, multi-role crew applying all core agentic AI patterns in a unified implementation.
- Implemented the **ReAct(Reasoning + Acting)** pattern enabling agents to reason step-by-step, dynamically **select tools** and **self-correct** before producing outputs, reducing hallucinations and improving task reliability.
- Applied the **Planning pattern** to separate deliberation from execution, having agents generate and validate a structured plan prior to acting and enabling **goal-directed behavior** across multi-step writing workflows.
- Designed **multi-agent crew** with specialized roles(Researcher, Writer and Editor) using hierarchical and sequential process modes with **inter-agent task-delegation** and context passing between crew members.
- Enforces **structured outputs** using **Pydantic** models to type-validate all agent responses, ensuring **schema consistency** and preventing downstream parsing failures across the crew pipeline.
- Implemented **guardrails** and **validation layers** to intercept malformed outputs, trigger **retry logic**, and enforce editorial constraints like word count bounds before output propagation.
- Added **Human-in-the-Loop(HITL)** controls at critical points, pausing execution to collect reviewer input and incorporate corrections before finalizing content.
- Managed **short-term, long-term, entity and contextual memory** across agent interactions, enabling the crew to retain context across tasks and reduce redundant information retrieval.

Certifications

- AWS Cloud Technical Essentials
- Big Data, Artificial Intelligence and Ethics

EDUCATION

Purdue University Northwest, Hammond, IN | Masters in Computer Science | May 2025